

Target Credentialing: Wnt Pathway & ERG/AR in Cancer

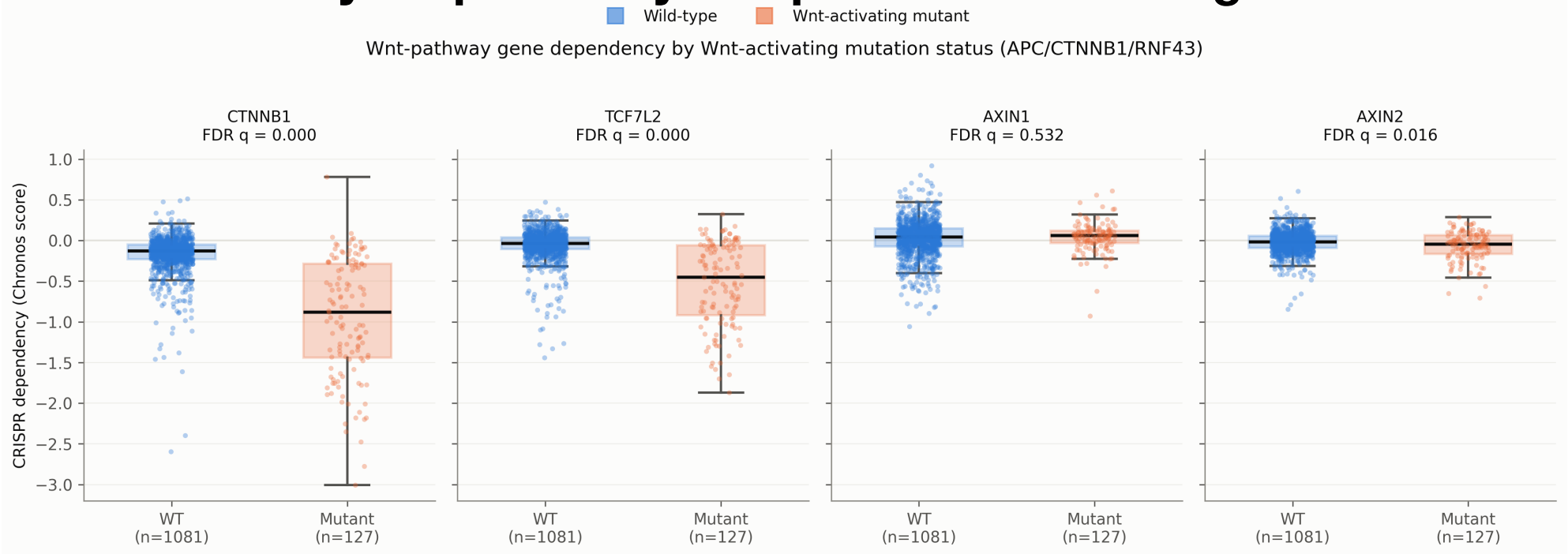
**Integrating functional and patient genomics to evaluate two
oncology target hypotheses**

Adrienne Yanez 14 Sep 2026

Background and Objective

- Objective: Evaluate two independent target hypotheses using orthogonal evidence - **functional genomics** (CRISPR dependency) and **patient genomics** (mutation/fusion prevalence and co-occurrence) to identify credentialed targets and indication expansion opportunities.
- **Wnt pathway:** Does Wnt-activating mutation status (APC/CTNNB1/RNF43) predict pathway dependency, and where does it matter clinically beyond CRC?
- **ERG/AR:** Does TMPRSS2-ERG fusion status track with AR-pathway activity in prostate cancer, potentially informing patient stratification?
- **Data sources:** DepMap (functional genomics); MSK-IMPACT and TCGA (patient genomics)

Wnt Pathway: Dependency and prevalence converge on CTNNB1

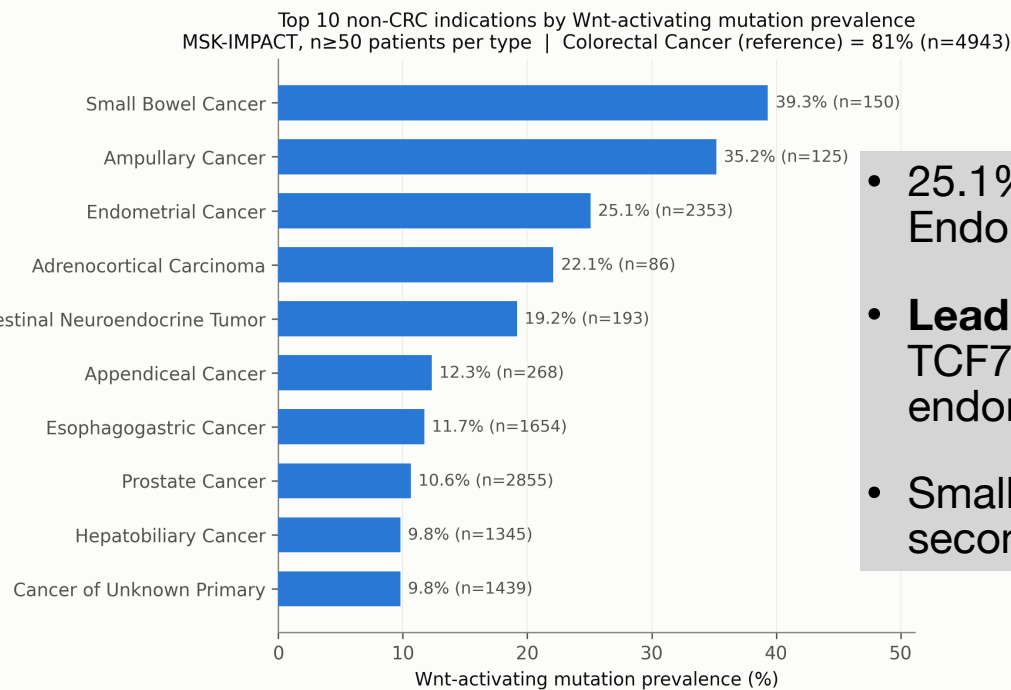


- CTNNB1 and TCF7L2 dependency strongly tracks Wnt-activating mutation status ($r=0.70$ and 0.57 ; $\text{FDR } q < 10^{-25}$)
- CTNNB1 is supported by both functional and patient genomics
- TCF7L2 is supported through the composite biomarker, not its mutation status (relevant to patient selection strategy)

Evidence Summary

Gene	Functional-genomics evidence	Patient-genomics evidence	Both
CTNNB1	Yes ($r=0.70$, $q \approx 10^{-37}$)	Yes, defines the Wnt-activating biomarker; literature-validated driver in endometrial cancer	Yes
TCF7L2	Yes ($r=0.57$, $q \approx 10^{-25}$)	Indirect, with dependency conferred by the APC/CTNNB1/RNF43 biomarker, not its own mutation status	Yes, via the biomarker
AXIN2	Weak ($r=0.14$, $q=0.016$)	Not separately assessed	Secondary
AXIN1	No ($q=0.53$)	Not separately assessed	Not supported

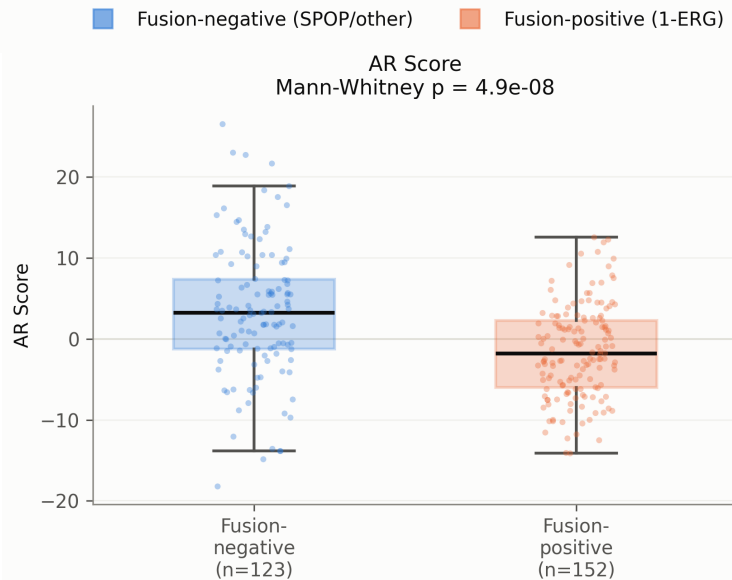
Indication Expansion: Endometrial Cancer



- 25.1% Wnt-activating mutation prevalence in Endometrial Cancer (n=2,353, literature-supported)
- **Lead indication-expansion hypothesis:** CTNNB1/TCF7L2 dependency in Wnt-activating-mutant endometrial cancer
- Small Bowel / Ampullary Cancer flagged as secondary, smaller-n candidates for follow-up

ERG/AR: A Stratification Signal for Combination Design

TMPRSS2-ERG fusion status vs. AR-pathway alteration (TCGA PRAD, n=275)



- ERG-fusion status associates with lower AR transcriptional signal (AR Score, $p=4.9 \times 10^{-8}$) but not AR-V7 status ($p=0.48$)
- Fusion status and AR activity appear to be **partially independent axes** (relevant to combination patient stratification)
- Evaluated on patient genomics alone; no usable public functional dependency data exists for this fusion

	Fusion-negative (n = 123)	Fusion-positive (n = 152)	Statistic
AR-V7 positive	26% (33)	23% (35)	Fisher's exact, $p = 0.48$

Recommended next steps

- Confirmatory functional work in endometrial cancer models
- Mechanistic follow-up on the ERG/AR Score relationship

Data sources:

- DepMap Public 26Q1 (depmap.org)
- MSK-IMPACT and TCGA Prostate Adenocarcinoma (Cell 2015), accessed via cBioPortal (cbioportal.org; Cerami et al. 2012, Gao et al. 2013)

Methods note: Developed using an AI-assisted, spec-driven workflow (Claude Code) with a persistent project specification guiding data validation and analytical decisions throughout. All data source substitutions and scope limitations are documented in the accompanying notebook.